

§ 13.7 triple integrals in Cartesian coordinates

$$\iiint_B f(x, y, z) dV = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(\bar{x}_k, \bar{y}_k, \bar{z}_k) \Delta V_k$$

$$\Sigma x \iiint_B xyz dV \quad B = [1, 2] \times [0, 2] \times [0, 3]$$

$$\int_1^2 \int_0^2 \int_0^3 xyz dz dy dx = \int_1^2 \int_0^2 xy \left[\frac{z^2}{2} \right]_0^3 dy dx = \int_1^2 \int_0^2 xy \cdot \frac{9}{2} dy dx$$

$$= \int_1^2 x \left[\frac{y^2}{2} \cdot \frac{9}{2} \right]_0^2 dx = \int_1^2 x \cdot 9 dx = \left[\frac{9x^2}{2} \right]_1^2 = \frac{9}{2} [4 - 1] = \frac{27}{2}$$

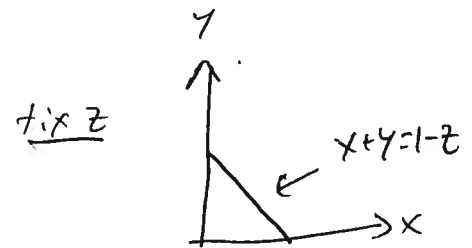
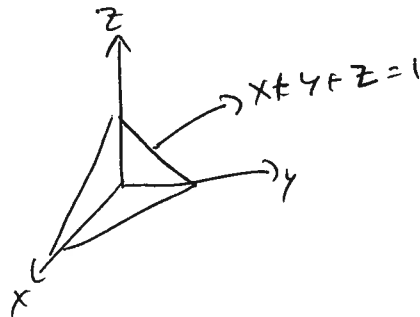
$$\Sigma x m = \iiint_S \rho dV \quad S = \text{pyramid with vertices } (0,0,0) (1,0,0) (0,1,0) (0,0,1)$$

$\rho = \text{constant}$

$$m = \int_0^1 \int_0^{1-z} \int_0^{1-z-y} \rho dx dy dz$$

$$= \int_0^1 \int_0^{1-z} \rho (1-z-y) dy dz = \int_0^1 \left[\rho \left(y - zy - \frac{y^2}{2} \right) \right]_0^{1-z} dz = \int_0^1 \rho \left[(1-z)^2 - \frac{(1-z)^2}{2} \right] dz$$

$$= \int_0^1 \rho \frac{(1-z)^2}{2} dz = -\rho \frac{(1-z)^3}{6} \Big|_0^1 = \frac{\rho}{6}$$



Ex Find the center of mass in the previous example

$$m \cdot \bar{z} = \iiint_S \rho \cdot z \, dV = \int_0^1 \int_0^{1-z} \int_0^{1-z-y} \rho \cdot z \, dx \, dy \, dz = \int_0^1 \rho \cdot z \frac{(1-z)^2}{2} \, dz$$

$$= \int_0^1 \frac{\rho}{2} (z^3 - 2z^2 + z) \, dz = \left[\frac{\rho}{2} \left(\frac{z^4}{4} - \frac{2z^3}{3} + \frac{z^2}{2} \right) \right]_0^1 = \frac{\rho}{2} \left(\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) =$$

$$= \frac{\rho}{2} \cdot \frac{1}{12}$$

$$\bar{z} = \frac{\rho}{24} / \frac{\rho}{6} = \frac{1}{4}$$

$$m \bar{x} = \iiint \rho x \, dV = \int_0^1 \int_0^{1-z} \int_0^{1-z-y} \rho x \, dx \, dy \, dz = \int_0^1 \int_0^{1-z} \rho \frac{x^2}{2} \Big|_0^{1-z-y} dy \, dz$$

$$= \int_0^1 \int_0^{1-z} \rho \frac{(1-y-z)^2}{2} dy \, dz = \int_0^1 \rho \frac{(1-y-z)^3}{6} \Big|_0^{1-z} dz =$$

$$= \int_0^1 -\frac{\rho}{6} (0 - (1-z)^3) dz = \int_0^1 \rho \frac{(1-z)^3}{6} dz = -\frac{\rho(1-z)^4}{24} \Big|_0^1 = \frac{\rho}{24}$$

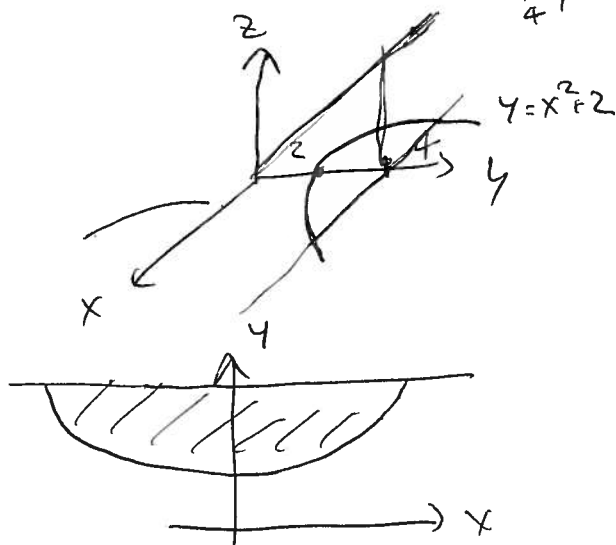
$$\bar{x} = 1/4 \quad (\text{Duh! symmetries!} \quad \bar{y} = 1/4)$$

Ex Find the volume of the solid bounded

by $y = x^2 + 2$, $y = 4$, $z = 0$ & $3y - 4z = 0$

$$\int_{-2}^2 \int_{x^2+2}^4 \int_0^{3/4 y} 1 \, dz \, dy \, dx$$

$$\int_{-2}^2 \int_{x^2+2}^4 \frac{3}{4} y \, dy \, dx = \int_{-2}^2 \left[\frac{3}{8} y^2 \right]_{x^2+2}^4 dx$$



$$= \int_{-2}^2 \frac{3}{8} y^2 \Big|_{x^2+2}^4 dx = \int_{-2}^2 \frac{3}{8} [16 - (x^2+2)^2] dx = \left[\frac{3}{8} \left[16x - \frac{x^5}{5} + \frac{4x^3}{3} + 4x \right] \right]_{-2}^2$$

$$= \frac{3}{8} \left[32 - \frac{32}{5} + \frac{32}{3} + 8 + 32 - \frac{32}{5} - \frac{32}{3} - 8 \right] = \frac{3}{8} \left[\frac{7 \cdot 64}{15} - 16 \right] = \frac{26}{5}$$

Ex Evaluate $\int_1^4 \int_{z-1}^{2z} \int_0^{4+2z} dx dy dz$, $\int_0^2 \int_1^2 \int_0^{\sqrt{xy}} zxyz dy dx dz$

$$\int_0^{\pi/2} \int_{\sin z}^0 \int_0^{2yz} \sin(x/y) dx dy dz$$

Ex Find the volume of the solid in the 1st octant bounded by $y = 2x^2$ & $y + 4z = 8$

Ex Find the volume of the solid in the 1st octant bounded by $y^2 + 64z^2 = 4$ & $y = x$.

Ex Find the center of mass of the part of the solid sphere $\{(x, y, z): x^2 + y^2 + z^2 \leq a^2\}$ that lies in the first octant. (Assume constant density ρ)